

Question 1:

Question 8 from section B

Let R be the ring $\mathbb{Z}[\sqrt{-13}]$

a)

$$A = \{s + t\sqrt{-13} : s, t \in \mathbb{Z}, s - t \equiv 0 \pmod{7}\}$$

As $s \equiv t \pmod{7}$ there exists an integer k such that,

$$s = t + 7k$$

Hence,

$$\begin{aligned} s + t\sqrt{-13} &= (t + 7k) + t\sqrt{-13} \\ &= t(1 + \sqrt{-13}) + 7k \end{aligned}$$

So

$$A \langle 7, 1 + \sqrt{-13} \rangle$$

By Definition 2.1 HB Chapter 17 p94,

To Show A is ideal, it must be closed under multiplication by any $r \in R$.
So we need the product of the generator and $\sqrt{-13}$ to stay in A .

$$\sqrt{-13}(1 + \sqrt{-13}) = \sqrt{-13} - 13$$

since

$$-13 \equiv 1 \pmod{7}$$

$$-13 = 1 - 14$$

$$\begin{aligned} \sqrt{-13} - 13 &= (\sqrt{-13} + 1) - 14 \\ &= (\sqrt{-13} + 1) + 7(-2) \end{aligned}$$

Since $(\sqrt{-13} + 1) \in A$ and $14 = 2(7) \in A$ it follows that A is closed under multiplication by any element of R and hence is an ideal.

b)

$$B = \{u + v\sqrt{-13} : u, v \in \mathbb{Z}, u + v \equiv 0 \pmod{7}\}$$

As $u = -v \pmod{7}$ there exists an integer k such that,

$$u = -v + 7k$$

Hence,

$$\begin{aligned} u + v\sqrt{-13} &= (-v + 7k) + v\sqrt{-13} \\ &= v(-1 + \sqrt{-13}) + 7k \end{aligned}$$

So

$$B\langle 7, -1 + \sqrt{-13} \rangle$$

because -1 is a unit in $\mathbb{Z}[\sqrt{-13}]$, therefore we can multiply the second generator by -1 and it will still generate the same ideal.

$$B\langle 7, 1 - \sqrt{-13} \rangle$$

Hence,

$$A\langle 7, 1 + \sqrt{-13} \rangle$$

and

$$B\langle 7, 1 - \sqrt{-13} \rangle$$

By Lemma 3.9 HB Chapter 17 p97, the product of two ideals A and B

$$AB\langle 7 \cdot 7, 7(1 - \sqrt{-13}), 7(1 + \sqrt{-13}), (1 + \sqrt{-13})(1 - \sqrt{-13}) \rangle$$

$$\langle 49, 7(1 - \sqrt{-13}), 7(1 + \sqrt{-13}), 14 \rangle$$

Since the $\text{hcf}(14, 49) = 7$ and $7(1 - \sqrt{-13})$ and $7(1 + \sqrt{-13})$ are multiples of 7 and,

$$7 = 49 - 3(14) \in AB$$

$$7 \subseteq AB$$

hence

$$AB\langle 7 \rangle$$